EMPLOYEE ATTRITION ANALYSIS

Business Report | IBM HR Analytics Dataset

1. Executive Summary

This report examines employee attrition patterns in a dataset of 1,470 employees across job roles, departments, age groups, income bands, and working conditions. The overall attrition rate is 16.1%. The strongest observed risk signals are overtime, early-career age, lower income, and specific frontline roles. These are associations in the available data—not proof that any single factor causes an employee to leave. The analysis combines Python-based exploratory analysis, PostgreSQL queries, predictive-model comparison, and a Power BI dashboard to support targeted retention decisions.

2. Business Problem Statement

Employee turnover can disrupt operational continuity, increase recruitment and onboarding effort, and place additional pressure on remaining employees. HR needs to understand where attrition is concentrated, which employee groups warrant closer review, and how to prioritize retention conversations. The business question is: Which workforce segments show the highest attrition, what measurable factors are associated with leaving, and how can HR use these findings to intervene earlier?

3. Key Findings & Analysis

- Overall attrition: 16.1% of the 1,470 employees in the dataset are recorded as having left.
- Overtime: Attrition is 30.5% among employees working overtime, compared with 10.4% among those not working overtime—about 2.9 times as high.
- Job roles: Sales Representatives have the highest observed role-level attrition (39.8%), followed by Laboratory Technicians (23.9%) and Human Resources employees (23.1%).
- Department: Sales has the highest department-level attrition (20.6%), followed by Human Resources (19.0%); Research & Development is lower at 13.8%.
- Age: Employees aged 18–25 show 35.8% attrition, the highest among the defined age bands; the rate is 19.1% for ages 26–35 and 9.2% for ages 36–45.
- Income: The project's income terciles show 26.1% attrition in the low-income group, versus 11.6% in the medium group and 10.6% in the high group.

These patterns suggest that retention efforts should first examine workload and overtime, early-career support, pay competitiveness, and role-specific working conditions. Results describe this dataset and should not be assumed to represent every organization or current workforce.

4. Root-Cause Assessment

The data points to several likely contributors that merit investigation rather than definitive causal conclusions. First, overtime is associated with substantially higher attrition, making workload distribution, staffing levels, and recovery time priority areas for review. Second, the higher attrition among lower-income employees may indicate pay competitiveness or perceived reward concerns; salary fairness should be tested within comparable roles and job levels. Third, high attrition in Sales Representative and Laboratory Technician roles suggests role-specific pressures, career progression, management support, or job expectations may be relevant. Finally, the elevated rate among employees aged 18–25 supports reviewing onboarding, training, early-career development, and manager check-ins. The dataset is observational and does not establish which mechanism directly causes departures.

5. Predictive Analysis & Limitations

Three classification approaches were evaluated on a stratified 80/20 train-test split. Logistic Regression achieved 75.2% accuracy, 61.7% recall, and 34.5% precision; Decision Tree achieved 71.8% accuracy, 48.9% recall, and 28.0% precision; Random Forest achieved 82.7% accuracy, 14.9% recall, and 38.9% precision. Logistic Regression is the most useful starting point if the priority is to identify more potential leavers, because its recall is highest; its low precision means many flagged employees may not leave. Random Forest's higher accuracy should not be interpreted as superior attrition detection because it misses most actual leavers. Predictions should be treated as prompts for supportive HR conversations, not as grounds for punitive or automated employment decisions.

6. Recommended Action Plan

1. Review overtime by team and role; investigate recurring excessive hours and rebalance staffing or workload where feasible.

2. Prioritize voluntary, confidential stay interviews for high-attrition roles and early-career employees; ask about workload, manager support, growth, and role expectations.

3. Audit compensation and progression within comparable roles and levels, especially for lower-paid groups; validate concerns before making pay changes.

4. Strengthen onboarding, mentoring, training, and career-path visibility for employees in their first years.

5. Monitor monthly attrition and leading indicators by role, department, age band, income band, and overtime status; evaluate whether interventions improve outcomes over time.

7. One-Line Business Solution

Implement a targeted, data-led retention programme that reduces avoidable overtime, reviews pay and career progression in high-risk roles, and provides early-career support—then track attrition trends to measure impact.

Scope & Method

Dataset: 1,470 employee records and 35 original columns. Data checks found no missing values or duplicate rows; three constant/low-value columns (EmployeeCount, Over18, StandardHours) were removed and analysis fields were engineered. Tools: Python (Pandas, visualization, scikit-learn), PostgreSQL (segmentation and prioritization queries), and Power BI (dashboard). Findings are descriptive associations; no causal experiment or external validation is shown in the supplied project outputs.